

# *Falls Prevention In The Elderly*

## **Out line**

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## **Introuduction**

Falls are a major health concern for older adults, posing risks of injury, decreased mobility, and a loss of independence. As people age, factors like muscle weakness, balance issues, vision problems, and chronic health conditions increase the likelihood of falling. In fact, falls are one of the leading causes of injury and hospitalization in elderly populations and can lead to serious complications such as fractures, head injuries, and a decline in overall health.

The impact of falls in the elderly goes beyond physical harm; falls can also affect psychological well-being, often leading to a fear of falling again. This fear can, in turn, reduce physical activity and social engagement, which may further weaken strength and balance and heighten the risk of future falls. Understanding the causes, prevention strategies, and treatment options for fall-related injuries is essential to supporting healthy aging and maintaining quality of life for older adults.

## **Definition Falling in elderly**

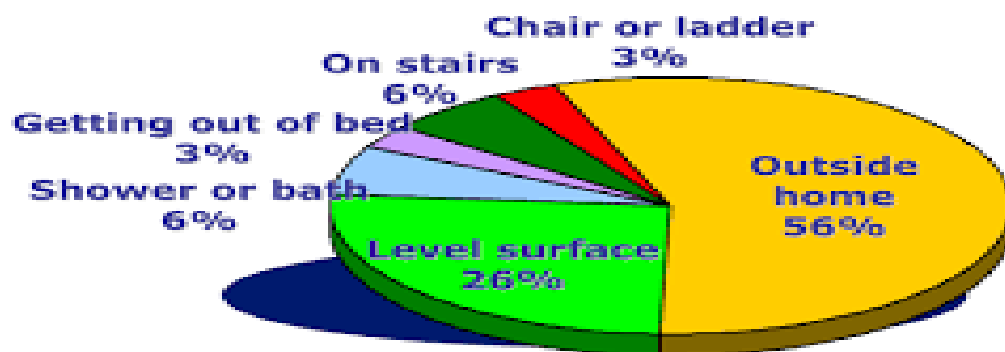
- WHO definition of a fall is an unexpected event in which a person coming to rest on the ground or another lower level with or without loss of consciousness or injury



## **Incidence of falling:**

- Falls are the second leading cause of accidental or unintentional injury deaths worldwide.
- Each year an estimated 646 000 individuals die from falls globally of which over 80% are in low- and middle-income countries.
- Adults older than 65 years of age suffer the greatest number of fatal falls.
- 37.3 million falls that are severe enough to require medical attention occur yearly

### **Distribution: Where**



## **Fall Etiology:**

Falls are usually multifactorial; intrinsic (internal), extrinsic (external) & acquired risk factors. It happens as a result of the interaction of intrinsic (personal) and extrinsic factors, some of these factors are modifiable (can be reduced) and others are not.

### **I-Intrinsic (Personal) factors for fall:**

1- Age; increase age >65.

2- Age related changes.

A- Sensory and balance:-

#### **Sensory:**

Includes (vision, inner ear, and proprioception)

- (vision): e.g., A decrease in visual acuity, visual field, night vision, color sensitivity, contrast sensitivity, depth perception & glare intolerance.
- (inner ear): helps eye stability and head orientation.

### **Balance:**

- (proprioception): A decline with age, → loss of vibratory sensation and joint position in lower extremities lead to increases in postural sway and impaired balance.

### **B-Musculoskeletal and gait:-**

**Musculoskeletal changes**; e.g., decline in muscle strength esp. in lower extremities, deterioration of cartilage in hips and knees & postural changes (Kyphosis, stooped posture), and arthritis.

**Gait**; aging has effects on gait e.g. decrease normal arm swing, hip and knee rotation, heel to floor. Decrease walking speed, shorter step length & narrow step width.

### **C-Behavioral/ psychological:**

e.g. fear of falling, threat of loss of independence, deny of frailty

- Anxiety & depression- lack of attention and awareness.

### **3- Medical condition:**

Chronic conditions including arthritis, stroke, incontinence, diabetes, Parkinson's, dementia.

### **4- Medications:**

Drug interaction and side effects e.g. sedative/ hypnotics, antihypertensive/ diuretics, analgesics, antidepressants, laxatives & alcohol

## II-Extrinsic or Environmental Factors:

### **1- Physical environment.**

**-Lack of stair handrails • Poor stair design • Lack of bathroom grab bars  
• Dim lighting or glare • Obstacles & tripping hazards • Slippery or uneven surfaces.**

### **2- Improper assistive device**

**e.g. Canes, Walkers, Wheelchairs - wrong height, improper use, ill design, ill repair& ill fit with environmental space.**

### **3- Footwear**

**e.g. high heel, improperly fitting shoes (loose shoes- foot problem, edema), slippery shoes, orthopedic shoes- heavy.**

## Another risk factors:

**Examples- Facility or hospital admission due to health change or decline (new environment), delirium due to illness, increased disability due to injury.**

## Cultural and Regional Considerations

**In some cultures, the use of assistive devices like canes or walkers is stigmatized, which may discourage individuals from adopting these life-saving tools. Additionally, access to fall prevention resources can vary by region, with low- and middle-income countries facing challenges like inadequate infrastructure, limited availability of safety devices, and lack of rehabilitation services.**

## Consequences of fall:

### **I. Physical consequences:**

**1- Mortality.**

**2- Morbidity.**

**3- Fractures.**

**\* Hip Fracture (pelvis, hip, femur) 90% of hip fractures are associated with falls in persons above 70 years.**

**\* Forearm (Wrist) Fracture.**

**\* Spine Fracture.**

**\* Ankle Fracture.**

**\* Upper arm, forearms, hand.**

**4- Soft tissue injuries.**

**5- Head trauma.**

**6- Joint distortions and dislocations.**

## **II. Psychological consequences:**

**- Fear of falling.**

**\* Restricted activity.**

**\* Loss of self-confidence.**

**\* Decrease in physical activity level and quality of life.**

**\* Fear of not being able to get up after a fall.**

### **Economic Impact of Falls**

**Beyond the physical and emotional toll, falls in older adults create a significant financial burden. Hospital stays, surgeries such as hip replacements, and prolonged rehabilitation can be expensive. For instance, fall-related injuries are one of the leading causes of healthcare spending in elderly populations. By preventing falls, we can not only improve quality of life but also reduce healthcare costs.**

### **Nursing interventions for safety and falls prevention:**

#### **Comprehensive Assessment Tools**

**Nurses use specific tools to assess an older adult's fall risk. For example, the Timed Up and Go Test measures mobility by timing how long it takes for someone to stand up, walk a short distance, and**

return. These tools allow us to identify at-risk individuals and create targeted interventions to prevent falls before they occur.

### **1. Intervention for predisposing factors:**

- 1. Identify factors (poor vision or hearing, unsteady gait) that may cause or contribute to falls to increase patient and family awareness.**
- 2. Encourage the patient to exercise regularly, especially weight-bearing exercises such as walking because exercise increases muscle strength, tone and balance.**
- 3. Instruct the patient to rise slowly in the morning and to sit on the side of the bed, with his feet dangling before getting out of bed because postural hypotension increases with age, especially if the patient is on diuretic or antihypertensive medications.**
- 4. Instruct the patient to avoid sudden movements of his head because aging causes changes in the vestibular system of the inner ear, which can lead to the loss of balance with sudden head movements.**
- 5. Educate patient about signs and symptoms of hypoglycemia and need to carry concentrated sugar.**
- 6. Regular eye examination**
- 7. Maintain adequate hydration and provide small frequent meals and rest periods after meal**

### **2. Environmental modification:**

- 1. Assess the patient's home for pets, children, small toys, loose rugs, and other small objects because the patient with poor vision may not be able to see small objects that could cause him to trip and fall.**
- 2. Teach the patient and his family about the need for safe light.**
  - Advise the patient to wear sunglasses to reduce glare.**



- Recommend the use of contrasting colors in household furniture, which enhance visual discrimination.
  - Install night-lights throughout the home.
3. Avoid wearing long clothes or bathrobes, especially while walking.
  4. Arrange furniture so that pathway are not obstructed.
  5. Provide stable furniture along pathway for balance and support .
  6. Tell the patient to remove all hazards from the stairs and hallways.
    - Install light switches that are easily accessible.
    - Install handrails that are easy to grasp from both sides of the stairs .
    - Painting the top and bottom steps by a bright color to recognize where and when the stairs begin and end.
  7. Keep Electrical wires bound together and tucked away so that they do not create a fall risk.
  8. In the bathroom, Use elevated toilet seat, use nonskid mat on tub floors and Provide handrails on both sides of bathroom
  9. Advise the patient to wear comfortable, properly fitting, and low-heeled shoes to provides stability and reduces the risk of falling.
  10. Tell the patient to watch for ramps, rough ground, and icy or wet pavements. To avoid busy streets, if possible; and to be especially careful when going out at night.
  11. Educate patient how to use assistive devices
  12. Advise the patient who lives alone to install an emergency call system, which allow him to call for help.
  13. While the patient is in the hospital,
    - Orient him to the environment.
    - Assess his ability to use the call bell, side rails, and bed positioning controls.
    - Keep the patient's bed at the lowest level and watch him closely at night to reduce the risk of falling.

### **3. Rehabilitation:**

**Adequate rehabilitation very important post fall**

- 1. Instruct the patient and his family to perform range-of-motion (ROM) exercises to prevent muscles contractures and restore joint mobility.**
- 2. Prevent prolonged immobility in the bedridden patient by frequent turning and ROM exercises. Prolonged immobility affects normal functioning of the joints and muscle**
- 3. Encourage the patient to perform ADLs, especially hair combing and dressing or walking to the bathroom to remain mobile and active.**
- 4. Instruct the patient in traction or immobilized by a cast to perform strength exercises to retain muscle tone of the immobilized body part.**
- 5. Encourage the patient to get out of bed, stand, and ambulate as soon as he's able or ordered.**
- 6. Assess the patient's need for assistive devices, such as a cane, walker or a wheelchair to be mobile and promotes independence.**
- 7. Avoid using physical restraints or keeping the bed's side rails up, unless necessary because restraints impair mobility and increase the patient's risk of falling.**
- 8. Instruct the patient on the correct way to fall to protect his head and face while falling and to call for help immediately.**

### **9. Psychological Support and Counseling**

**The fear of falling can be debilitating, leading to reduced physical activity and increased isolation. Psychological interventions like cognitive-behavioral therapy can help older adults regain confidence, while group exercise programs like Tai Chi can provide both physical and emotional support to reduce fall anxiety**

### **Bone Health and Fracture Prevention**

**Bone health is closely tied to the severity of injuries from falls. Older adults with osteoporosis are more likely to suffer fractures. Ensuring adequate calcium and Vitamin D intake, combined with regular weight-bearing exercises, can strengthen bones and reduce the risk of fractures, particularly .hip fractures which are often life-changing**

